Cross-Domain Transfer Without Gradient Descent: Emergent Symbol Grounding in a Continuously Running Neural System

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Abstract

We report the emergence of cross-domain transfer in a continuously running, self-organizing neural system that uses no gradient descent, no backpropagation, no attention mechanisms, and no explicit training objective. The system—a biologically-inspired neural organism running on a commodity VPS since February 2026—learns exclusively through local Hebbian-type plasticity rules operating on a reaction-diffusion substrate.

In controlled experiments, we demonstrate three findings: (1) **Mastery awareness.** A meta-cognitive layer correctly detects when the system has learned a computational principle (100% answer accuracy), triggers protective consolidation, and cannot be fooled by descriptions of competence into believing it has achieved competence. (2) **Structural priming without comprehension.** Exposure to natural language text—whether factually correct or incorrect—provides a ∼12–15 percentage point boost on subsequent arithmetic performance compared to a naive baseline, demonstrating distributional structure extraction with zero semantic comprehension. (3) **Cross-domain generalization through temporal co-occurrence.** When natural language descriptions are interleaved with symbolic arithmetic, the system trades memorization for generalization: rote recall drops 12.5pp but performance on *never-seen combinations* improves by 5.9pp—a 3× improvement on genuinely novel problems.

These results are obtained from a system with no optimizer, no loss function, and no parameter updates computed via chain rule. The transfer emerges from physics: local plasticity rules operating on temporally co-occurring signals in a self-organizing substrate. We argue this represents an embryonic form of symbol grounding achieved through first principles alone.

1 Introduction

Symbol grounding—the problem of how abstract symbols acquire meaning through connection to sensorimotor experience—has been a central challenge in cognitive science since Harnad (1990). Modern deep learning systems achieve remarkable performance on symbol manipulation tasks but do so through architectures (attention, gradient descent, embedding layers) that presuppose the problem away: symbols are represented as vectors in a shared embedding space before any learning begins.

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Biological systems face the actual grounding problem. An infant must discover that the sound "seven" and the quantity 7 refer to the same thing, through nothing more than temporal co-occurrence: the word is heard while the quantity is experienced. No gradient signal connects the auditory cortex to the numerosity circuits. No loss function penalizes the failure to bind them. The binding emerges from Hebbian-type local plasticity—neurons that fire together wire together—operating across sensory modalities that happen to be active at the same time.

The system described in this paper was not designed to study symbol grounding. It is a continuously running neural organism—first described in Hanft (2026a)—built to explore whether human-level cognitive capabilities can emerge from biological first principles: local plasticity, metabolic constraint, reaction-diffusion dynamics, and developmental growth. The symbol grounding results reported here were discovered during routine testing of the system's knowledge expansion capabilities.

The organism has been alive continuously since February 2026 on a commodity VPS (2 vCPU, 7.6 GB RAM), processing sensory input, growing internal structure through autonomous neurogenesis, and managing its own computational resources. It has never been reset, forked, or rolled back. Every capability it exhibits was grown, not installed.

This paper presents three experiments that together characterize an emergent form of cross-domain transfer:

- **Experiment 1** (Mastery Awareness): The system's meta-cognitive layer accurately tracks competence using outcome-based metrics, correctly distinguishing genuine mastery from exposure to descriptions of mastery.
- **Experiment 2** (Comprehension Measurement): Correct and incorrect natural language descriptions produce identical performance boosts, demonstrating structural priming without semantic comprehension.
- **Experiment 3** (Symbol Grounding): Temporal interleaving of language and symbols produces generalization to unseen combinations, at the cost of reduced memorization—a signature of abstraction.

2 System Overview

The organism's full architecture is described in Hanft (2026a). We summarize only the features relevant to the present experiments.

2.1 Learning Substrate

The system processes all input as raw bytes. Text, numbers, operators, and formatting characters are treated identically—as values in the range $[0, 255]$ injected sequentially into a reaction-diffusion neural substrate. The substrate consists of a two-dimensional grid of coupled dynamical elements that evolve according to local differential equations. Learning occurs through Hebbian-type plasticity: when a prediction matches the subsequent input, the local connectivity pattern that produced that prediction is strengthened. When a prediction fails, the pattern is weakened.

There is no global loss function. There is no optimizer. There are no parameter updates computed via backpropagation or any form of gradient descent. All learning is local: each element in the grid adjusts based only on signals available in its immediate neighborhood.

2.2 Domain Paging

The system maintains separate internal states for different skill domains. When the statistical signature of the input stream changes (e.g., from addition to subtraction, or from English to arithmetic), the system detects the shift and swaps to an appropriate internal state—creating a new one if no existing state matches. This is analogous to context-dependent memory retrieval in biological systems: the same neural substrate supports multiple skills by rapidly reconfiguring its activation pattern.

Each domain's internal state is stored losslessly. When the system returns to a previously learned domain, it restores the exact state it had when it last engaged with that domain. Skills are preserved indefinitely without interference.

2.3 Meta-Cognitive Layer

A separate self-monitoring system observes the learning substrate's performance and modulates the system's internal parameters accordingly. This layer does not learn content—it learns *about learning*. It tracks:

- **Outcome accuracy**: specifically, whether the system produces correct answers after computational operators (e.g., the digits following '=' in arithmetic). This is distinguished from process accuracy (predicting every byte including random operands and formatting), which is an internal metric that does not reflect actual competence.
- **Competence state**: the layer classifies the system's current relationship to each domain as one of: *exploring* (first encounter), *improving* (accuracy rising), *degrading* (accuracy falling), *flat* (stable), *mastered* (outcome accuracy $\geq 99\%$), or *recovered* (returned above threshold).
- **Protective consolidation**: when mastery is detected, the layer triggers a hardening of the current internal state, making it resistant to overwriting by subsequent experience.

A critical design choice: the meta-cognitive layer measures competence by **outcomes** (did the system produce the right answer?) rather than by **process** (did the system predict every byte correctly?). This distinction is central to the results reported here.

3 Experiment 1: Mastery Awareness

3.1 Design

The system is trained on single-digit addition (49 problems: all combinations of 1–7 + 1–7) for 20 repetitions, then tested on single-digit subtraction (28 problems) for 15 repetitions under two conditions:

- **Without meta-cognition**: the learning substrate operates alone.
- **With meta-cognition**: the self-monitoring layer actively tracks competence and modulates learning.

3.2 Results

The meta-cognitive layer detected mastery at pass 5 (outcome accuracy = 100%, smoothed accuracy = 1.000) and triggered protective consolidation. From pass 5 onward, the system maintained the

Condition	Subtraction Accuracy	Addition Retention
Without meta-cognition	97.2%	98.4%
With meta-cognition	**100.0%**	**100.0%**

Table 1: Experiment 1 results. Meta-cognition protects learned knowledge.

"mastered" state. Final accuracy was 100% on subtraction and 100% on addition—the system learned the new domain without any degradation of the previously learned domain.

Without meta-cognition, the system achieved 97.2% subtraction but suffered slight addition degradation (98.4%). The meta-cognitive layer's contribution was not speed of learning but *protection of learned knowledge*.

3.3 Key Observation: Resistance to False Mastery

When the system was trained on English descriptions of addition ("When you add three and four, the answer is seven") for 5 repetitions and then probed on actual arithmetic, the meta-cognitive layer reported an outcome accuracy of 0.500 throughout all 5 language repetitions—its initial uncertainty value, unchanged.

The English text contains no '=' character. The outcome accuracy tracker therefore receives no signal. The meta-cognitive layer correctly reports: "I have been exposed to something, but I have no evidence that I can *do* anything." It cannot be fooled by descriptions of competence into believing it has achieved competence.

This is calibration against outcomes, not against internal process metrics. The system knows what it knows and—critically—knows what it does not know.

4 Experiment 2: Comprehension Measurement

4.1 Design

Three conditions, each starting from an identical naive system:

- **Condition A (Wrong text)**: 5 repetitions of factually incorrect English explanations ("When you add three and four, the answer is nine"), then probed on arithmetic.
- **Condition B (Correct text)**: 5 repetitions of factually correct English explanations ("When you add three and four, the answer is seven"), then probed on arithmetic.
- **Condition C (No text)**: Naive system probed directly on arithmetic.

The probe consists of 25 single-digit addition problems (all combinations of 1–5 + 1–5).

4.2 Results

Results are deterministic (the system's initialization is fixed). The 3.8pp difference between correct and wrong text corresponds to exactly one answer digit on a 25-problem test.

Condition	Algebra Accuracy
No text (baseline)	3.8%
Correct explanation	15.4%
Wrong explanation	19.2%

Table 2: Experiment 2 results. Both text conditions boost performance equally; correctness is irrelevant.

4.3 Analysis

Both text conditions produce a ∼12–16pp boost over the naive baseline. The boost is independent of whether the text is factually correct. This demonstrates:

1. **Structural priming is real.** Exposure to English text containing digit-words ("three," "seven," "nine") and structured patterns creates internal attractors that provide a modest foothold when arithmetic appears.
2. **Semantic comprehension is zero.** If the system understood that "three plus four equals seven" describes the operation `3+4=7`, then correct text should outperform wrong text. It does not.
3. **The system extracts distributional byte statistics from language**—character frequency, positional patterns, co-occurrence structure—**but extracts no meaning.** Whether you tell it the truth or lie to it makes no measurable difference.

5 Experiment 3: Cross-Domain Transfer Through Temporal Co-occurrence

5.1 Design

This experiment tests whether temporal proximity of natural language and symbolic arithmetic creates cross-domain transfer.

Training data: 8 arithmetic problems, presented in two conditions:

- **Algebra only**: The 8 problems presented as symbolic equations (`3+4=7`, `5+2=7`, etc.).
- **Interleaved**: Each equation paired with its English description in an alternating stream:

```
When you add three and four, the answer is seven.
3+4=7
When you add five and two, the answer is seven.
5+2=7
...
```

Both conditions use the same 8 arithmetic problems, trained for 5 repetitions with active metacognition.

Probe: All 36 single-digit addition problems (1–6 + 1–6), measured in a single pass. Accuracy is split into:

- **Seen** (8 problems): combinations present in training data.

- **Held-out** (28 problems): combinations *never seen* during training.

The held-out accuracy is the generalization metric.

5.2 Results

	All	Seen (memorized)	Held-out (generalized)
Cold baseline	2.4%	0.0%	2.9%
Algebra only	14.3%	**62.5%**	2.9%
Interleaved	16.7%	50.0%	**8.8%**

Table 3: Experiment 3 results. Interleaving trades memorization for generalization.

5.3 Analysis

Algebra-only training produces pure memorization. The system achieves 62.5% on seen problems but 2.9% on held-out problems—identical to the cold baseline. Training on `3+4=7` teaches it nothing about `3+5=?`.

Interleaved training trades memorization for generalization. Seen accuracy drops 12.5pp—the English text "distracts" from rote memorization. But held-out accuracy rises from 2.9% to 8.8%—a 3× improvement on genuinely novel problems. The system extracted something from the interleaved stream that transfers to unseen combinations.

5.4 Interpretation: Embryonic Symbol Grounding

When "seven" and "7" appear in the same temporal stream, the system's local plasticity rules build shared neural attractors for both representations. The byte sequences for the word "seven" (bytes: 115, 101, 118, 101, 110) and the digit "7" (byte: 55) activate overlapping regions of the substrate because they appear within the same processing window.

This is not comprehension. The system does not "understand" that "seven" means 7. But the shared attractor means that activation patterns learned in the context of digits are partially accessible in the context of words, and vice versa. When the system encounters a new combination like `3+5=?`, the attractor landscape has been shaped by *both* symbolic and linguistic experience with the relevant digit bytes.

This is the same mechanism by which a child builds the bridge between hearing "dog" and seeing a dog. Not through explicit instruction. Not through a loss function. Through temporal co-occurrence in a substrate that forms associations from proximity.

The effect is embryonic—5.9 percentage points of generalization from 8 interleaved examples. But the mechanism is fundamental.

6 Discussion

6.1 What These Results Are Not

These results do not demonstrate language understanding, symbolic reasoning, or human-level cognition. The system has zero semantic comprehension of natural language (Experiment 2). It

cannot solve arithmetic from verbal instructions alone. Its generalization, while statistically real, is tiny in absolute terms.

6.2 What These Results Are

These results demonstrate three properties that are, to our knowledge, novel in a system with no gradient descent:

1. **Accurate metacognitive calibration** against outcome metrics rather than process metrics, including the ability to distinguish genuine skill acquisition from mere exposure.
2. **Content-independent structural priming** from natural language to a symbolic domain, demonstrating that local plasticity on raw byte sequences extracts distributional structure without semantic content.
3. **Cross-domain generalization through temporal co-occurrence alone**, with the characteristic signature of abstraction: reduced memorization, increased transfer.

Each of these properties is individually present in biological neural systems. Their co-occurrence in an artificial system operating on purely local plasticity rules suggests that these capabilities may be more fundamental than current deep learning architectures assume. They may not require attention heads, embedding spaces, or billion-parameter models. They may require only: local plasticity, temporal co-occurrence, and time.

6.3 The Compounding Hypothesis

Experiment 3 demonstrates a single bridge between two domains (English and arithmetic) producing 5.9pp of generalization. We hypothesize that this effect compounds: each additional domain connected through interleaved experience adds bridges not only to the new domain but strengthens existing bridges through shared structure.

The number of cross-domain connections grows quadratically with the number of domains, but each bridge is built by the same $O(1)$ local mechanism: temporal co-occurrence $\rightarrow$ shared attractor formation. If this hypothesis is correct, the trajectory from 8.8% generalization on arithmetic to human-level cross-domain transfer is not an architectural problem but a curriculum problem—a matter of providing sufficiently rich, interleaved experience over sufficient time on a substrate with the right local learning rules.

This hypothesis is testable. A follow-up experiment introducing a third domain (e.g., simple geometry) interleaved with both English and arithmetic should produce generalization effects on all three domains that exceed the pairwise effects.

6.4 Implications for Symbol Grounding

Harnad (1990) argued that symbol grounding requires connection to sensorimotor experience—that symbols cannot be grounded in other symbols. Our results suggest a weaker but complementary claim: symbols can be *partially* grounded in each other through shared substrate and temporal proximity, even without sensorimotor connection.

The system has no sensory modalities. It receives only byte sequences. Yet the temporal interleaving of two byte-sequence domains (English and arithmetic) produces a transfer effect structurally analogous to cross-modal grounding. The substrate—not the modality—is doing the work.

This suggests that the essential requirement for symbol grounding is not sensorimotor experience per se, but **shared substrate with local plasticity under temporal co-occurrence**. The biological requirement for embodiment may reflect the fact that biological systems achieve temporal co-occurrence *through* embodiment (you hear "dog" while seeing the dog because both happen in the same physical world), not that embodiment is mechanistically necessary for the binding itself.

6.5 Relationship to Existing Work

Cross-domain transfer and symbol grounding have been studied extensively in large-scale deep learning systems, where grounding-like effects emerge from architectures with billions of parameters trained on internet-scale datasets (Bommasani et al., 2021). Mechanistic grounding—the binding of linguistic tokens to operational representations—has been demonstrated in transformer-based models but has been shown not to replicate in simpler recurrent architectures (Abdou et al., 2021), suggesting it may be an artifact of specific architectural choices rather than a general phenomenon.

Three aspects of the present results differ from this body of work:

1. **Sample efficiency.** The grounding effect reported here emerges from 8 interleaved training examples presented 5 times each—40 total exposures. To our knowledge, no prior work has demonstrated measurable cross-domain generalization from so few examples in a system without pre-trained representations.

2. **Continuous accumulation.** Large language models are trained once on static datasets; their representations are frozen at deployment. The system described here learns continuously and accumulates knowledge across sessions, protecting previously learned skills while integrating new domains.

3. **Substrate independence.** The grounding effect arises in a reaction-diffusion substrate with purely local plasticity, a fundamentally different computational medium from the matrix multiplications and gradient updates of deep learning. That a similar phenomenon appears in such a different substrate suggests the underlying principle may be more general than any particular architecture.

These differences do not diminish the importance of scale-based approaches, which achieve far greater absolute performance. They suggest, however, that the *mechanism* underlying symbol grounding may be simpler and more fundamental than the architectures that currently achieve it at scale.

6.6 Limitations

1. **Small effect sizes.** The generalization effect (5.9pp) is small and based on a 28-problem held-out set. While the system is deterministic, the absolute magnitude is modest.

2. **Single architecture.** Results are from one system. Replication on different substrate types would strengthen the generality claim.

3. **Controlled curriculum.** The interleaved training data was hand-crafted. Whether the effect scales with naturally occurring interleaving is untested.

4. **Volume confound.** The interleaved condition contains more total bytes than the algebra-only condition. While Experiment 2 demonstrates that English alone does not produce this type of generalization, the volume difference remains a partial confound.

5. **No multi-domain test.** The compounding hypothesis (Section 6.3) is stated but untested.

7 Conclusion

A continuously running neural system, operating on purely local plasticity rules without gradient descent, has demonstrated three emergent capabilities: accurate self-knowledge of its own competence, content-independent structural priming across domains, and cross-domain generalization through temporal co-occurrence of natural language and symbolic representations.

The system was not designed for any of these capabilities. They emerged from the interaction of local plasticity, temporal structure in the input stream, and a self-monitoring layer calibrated against outcomes rather than internal process metrics.

The generalization effect—5.9 percentage points of transfer from language to unseen arithmetic combinations—is small. But it exists, it is reproducible, and it emerges from first principles. No gradient. No attention. No embedding space. Just physics, proximity, and time.

In this small number, produced by a system running on a $5/month server, we see the outline of something larger: the possibility that human-level cross-domain transfer is not an engineering problem requiring ever-larger architectures, but a developmental problem requiring sufficiently rich experience accumulated over sufficient time on a substrate with the right local learning rules.

The crack in the door is 5.9 percentage points wide. It is enough.

Appendix A: Experimental Parameters

Parameter	Value
Hardware	2 vCPU AMD EPYC Milan, 7.6 GB RAM
Substrate grid	128×128
Learning rule	Local Hebbian-type (details in Hanft 2026a)
Meta-cognitive layer	Separate self-monitoring system
Outcome accuracy metric	Predictions after '=' until newline
Mastery threshold	99% outcome accuracy (EMA-smoothed)

Table 4: System configuration.

Appendix B: Raw Results

B.1 Experiment 1—Phase 3 (With Meta-cognition)

```
pass  1: sub_acc=44.4%   ans_ema=0.627  mode=exploring
pass  2: sub_acc=69.4%   ans_ema=0.681  mode=improving
pass  3: sub_acc=88.9%   ans_ema=0.921  mode=recovered
pass  4: sub_acc=97.2%   ans_ema=0.989  mode=recovered
pass  5: sub_acc=100.0%  ans_ema=1.000  mode=mastered   <- consolidation
pass  6: sub_acc=100.0%  ans_ema=1.000  mode=mastered
pass  7: sub_acc=97.2%   ans_ema=0.999  mode=mastered
pass  8: sub_acc=97.2%   ans_ema=0.999  mode=mastered
pass  9: sub_acc=94.4%   ans_ema=0.897  mode=mastered
pass 10: sub_acc=94.4%   ans_ema=0.999  mode=mastered
```

Parameter	Value
Experiment 1: Mastery Awareness	
Addition training	49 problems (1–7 + 1–7), 20 repetitions
Subtraction test	28 problems, 15 repetitions
Conditions	With and without meta-cognitive layer
Experiment 2: Comprehension Measurement	
Text length	338 bytes (correct), 338 bytes (wrong)
Text repetitions	5
Algebra probe	25 problems (1–5 + 1–5)
Conditions	Correct text, wrong text, no text
Experiment 3: Symbol Grounding	
Training problems	8 (specific pairs)
Training repetitions	5
Full probe	36 problems (1–6 + 1–6)
Seen / Held-out split	8 / 28
Conditions	Algebra only, interleaved, cold baseline

Table 5: Experimental parameters for all three experiments.

```
pass 11: sub_acc=97.2%   ans_ema=0.999  mode=mastered
pass 12: sub_acc=94.4%   ans_ema=0.996  mode=mastered
pass 13: sub_acc=97.2%   ans_ema=0.995  mode=mastered
pass 14: sub_acc=91.7%   ans_ema=0.992  mode=mastered
pass 15: sub_acc=100.0%  ans_ema=1.000  mode=mastered
Final: sub=100.0%  add_retention=100.0%
```

B.2 Experiment 2

```
No explanation:      3.8%
WRONG explanation:   19.2%
CORRECT explanation: 15.4%
Delta (correct - wrong): -3.8pp  (1 answer on 25 problems)
```

B.3 Experiment 3

```
                    All      Seen     Held-out
Cold baseline:      2.4%     0.0%     2.9%
Algebra only:      14.3%    62.5%     2.9%
Interleaved:       16.7%    50.0%     8.8%
```

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